PLANTNEST - PLANT CARE AND REMINDER SYSTEM

A MINI PROJECT REPORT

IT2305 – FULL STACK WEB DEVELOPMENT LABORATORY

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DEPARTMENT OF INFORMATION TECHNOLOGY

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BONAFIDE CERTIFICATE

Certified that the project report **"PLANTNEST - PLANT CARE AND REMINDER SYSTEM "** is the bonafide work of **" JABEEN A (920424205033) , SHAKTHI PRIYA TG (920424205094) & YALINIDEVI T (920424205112)"** who carried out the mini project work as a part of **IT2305 – Full Stack Web Development Laboratory** under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

CHAPTER 1

INTRODUCTION

1.1 MERN Stack

MERN Stack is a full-stack JavaScript technology stack used for developing modern web applications. MERN stands for MongoDB, Express.js, React, and Node.js. These technologies work together to support the development of the database, backend services, and frontend interface of a web application.

The PlantNest project is developed using the MERN Stack as a **Plant Care and Reminder System**. The repository contains separate `client` and `server` components, providing a structured approach for developing the frontend and backend of the application.

1.2 Importance of MERN Stack

MERN Stack provides a convenient approach for developing full-stack web applications using JavaScript-based technologies. It allows developers to create interactive frontend interfaces, backend services, and database-driven applications using a connected technology stack.

For an application such as PlantNest, MERN Stack helps organize user information, plant details, watering schedules, and reminder-related operations. It also provides a flexible structure that can be extended with additional plant care features in the future.

1.3 Basic Terminologies

• **MongoDB:** A NoSQL database used to store application information in document-based structures.

• **Express.js:** A backend web framework for Node.js used to develop server-side applications and APIs.

• **React:** A JavaScript library used to create interactive and component-based user interfaces.

• **Node.js:** A JavaScript runtime environment used to execute JavaScript on the server side.

1.4 Components of MERN Stack

1.4.1 MongoDB

MongoDB is a NoSQL database that stores information using flexible document-based structures. In the PlantNest application, MongoDB is used as the database environment for storing application data. The project instructions specifically require MongoDB to be started locally or through MongoDB Atlas.

1.4.2 Express.js

Express.js is a lightweight backend framework that works with Node.js. It is used to handle server-side requests, create API routes, and connect the frontend application with backend functionality.

1.4.3 React

React is a JavaScript library used to develop dynamic and interactive user interfaces. It supports component-based development and helps in creating reusable parts of the application interface.

1.4.4 Node.js

Node.js is a JavaScript runtime environment that allows JavaScript code to execute outside the browser. It is used to develop the server-side part of the PlantNest application and works with Express.js to provide backend services.

1.5 Operations in MERN Stack

Common operations in a MERN Stack application include:

• **Create:** Adding new information such as a new plant record.

• **Read:** Retrieving and displaying stored plant or user information.

• **Update:** Modifying existing plant or application information.

• **Delete:** Removing unwanted plant records.

• **API Communication:** Transferring information between the React frontend and the backend services.

In PlantNest, the backend provides API operations for authentication and plant management, including adding, retrieving, updating and deleting plants.

1.6 Need for MERN Stack

MERN Stack provides:

• Full-stack web development using JavaScript

• Interactive and component-based user interfaces

• Flexible database management using MongoDB

• Efficient backend development using Node.js and Express.js

• Easy communication between frontend and backend

• A scalable structure for developing web applications

1.7 Applications of MERN Stack

MERN Stack can be used to develop:

• Plant care and management applications

• E-commerce applications

• Online booking systems

• Social networking applications

• Content management systems

• Business management applications

• Real-time web applications

CHAPTER 2

METHODOLOGY

2.1 Objective

The main objective of the **PlantNest – Plant Care and Reminder System** is to help users manage their plants and maintain regular watering schedules through a web-based application.

The system allows users to register and log in, add and manage plant records, define watering intervals, and keep track of upcoming watering dates. When a plant is marked as watered, the system recalculates the next watering date based on the specified watering interval.

The application also provides a daily reminder mechanism to help users remember plants that require watering. By automating these activities, PlantNest aims to reduce the possibility of forgetting plant care and provide a convenient way to manage multiple plants.

2.2 Problem Statement

Taking care of plants requires regular watering and proper maintenance. People who have multiple plants may find it difficult to remember the individual watering schedules of each plant. Since different plants may require watering at different intervals, manually maintaining these schedules can become inconvenient.

Forgetting to water plants at the required time may affect their growth and overall health. Traditional methods such as maintaining notes or remembering watering dates manually may not always be reliable.

The **PlantNest – Plant Care and Reminder System** is proposed to address these problems by providing an online platform where users can maintain plant information, set watering intervals, update watering status, and receive reminders for plants that require attention.

2.3 Project Description

The **PlantNest – Plant Care and Reminder System** is a full-stack web application designed to help users manage their plants and watering schedules in an organized way.

User Functionalities

• **User Registration and Login:** Users can create an account and log in to access the PlantNest application. The project uses JWT authentication for registration and login.

• **Plant Management:** Users can create and manage plant records through the application.

• **Plant Details:** Users can maintain information about their plants and their watering requirements.

• **Watering Interval:** Users can specify how frequently a particular plant needs to be watered.

• **Watered Function:** When the user marks a plant as watered, the system calculates the next watering date based on the configured interval.

• **Plant CRUD Operations:** The system supports adding, viewing, updating and deleting plant records.

• **Due Plant Tracking:** The backend provides a facility to retrieve plants that are due for watering.

• **Email Reminder:** The application includes a daily 8 AM cron-based email reminder system for plants that require attention. If SMTP is not configured, the repository specifies that the reminder is logged to the console.

Application Structure

The project is organized into two main parts: `client` and `server`. The `client` contains the frontend application, while the `server` contains the backend functionality and API services.

The backend can be started using the provided npm development command, while the frontend runs separately using its own development command. The repository specifies `http://localhost:5173` as the frontend development address.

CHAPTER 3

RESULTS AND DISCUSSION

Screenshots

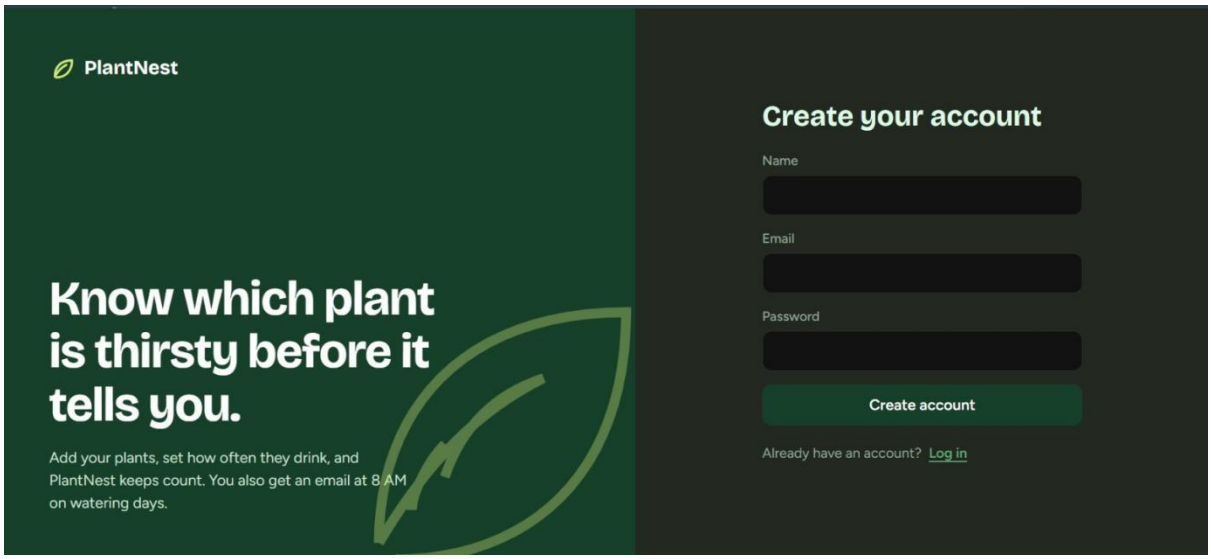


FIGURE 3.1: PlantNest User Registration Page

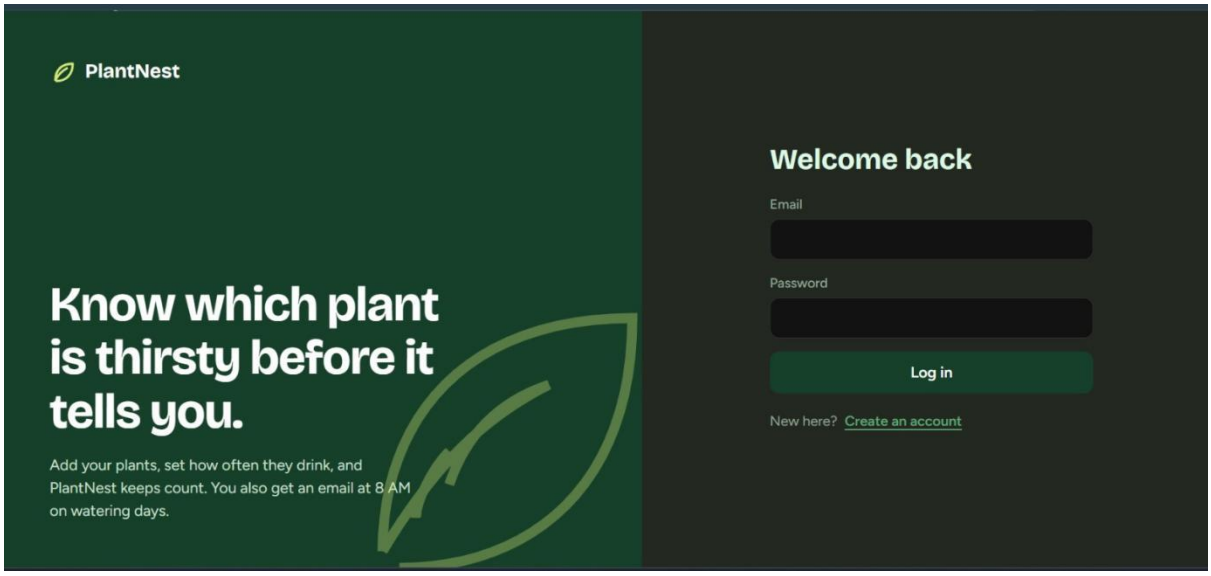


FIGURE 3.2: PlantNest User Login Page

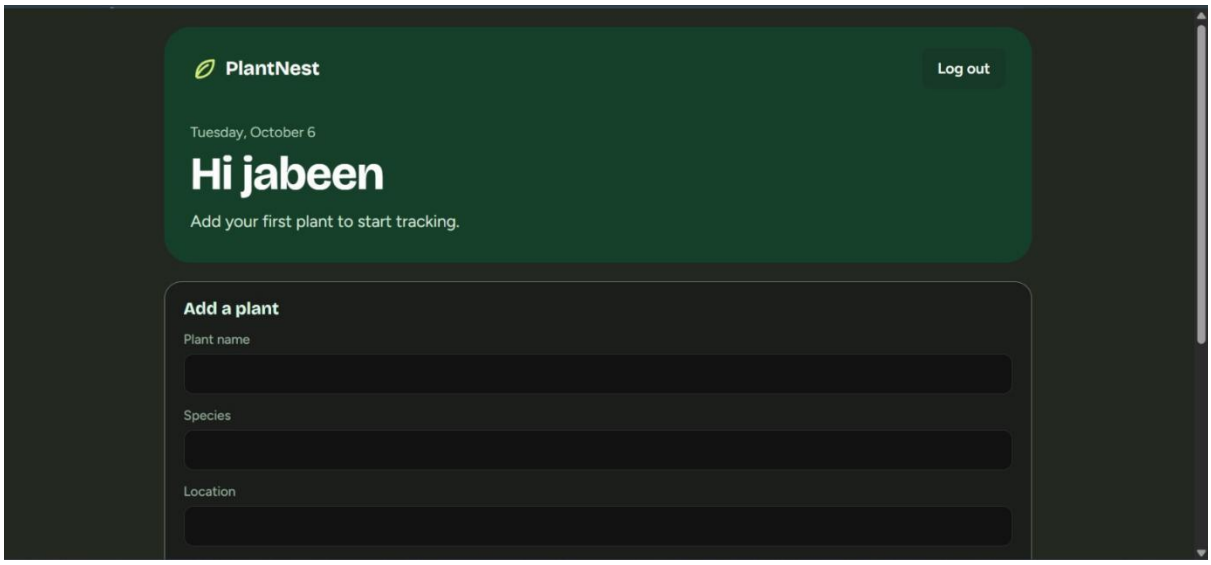


Figure 4.3 – PlantNest Plant Dashboard

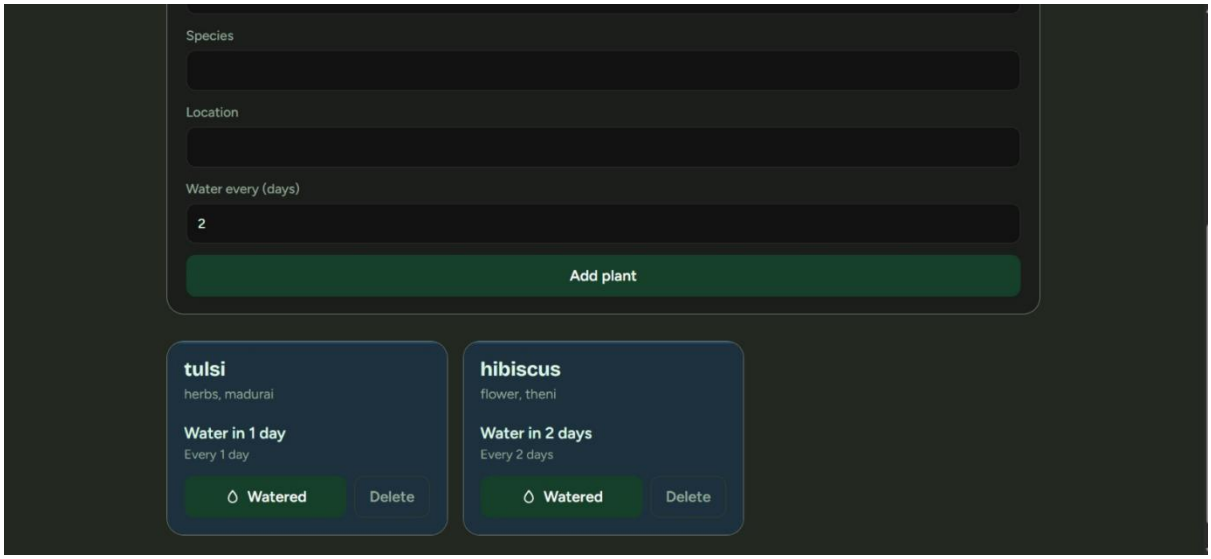


Figure 4.4 – PlantNest Plant Tracking and Watering Schedule

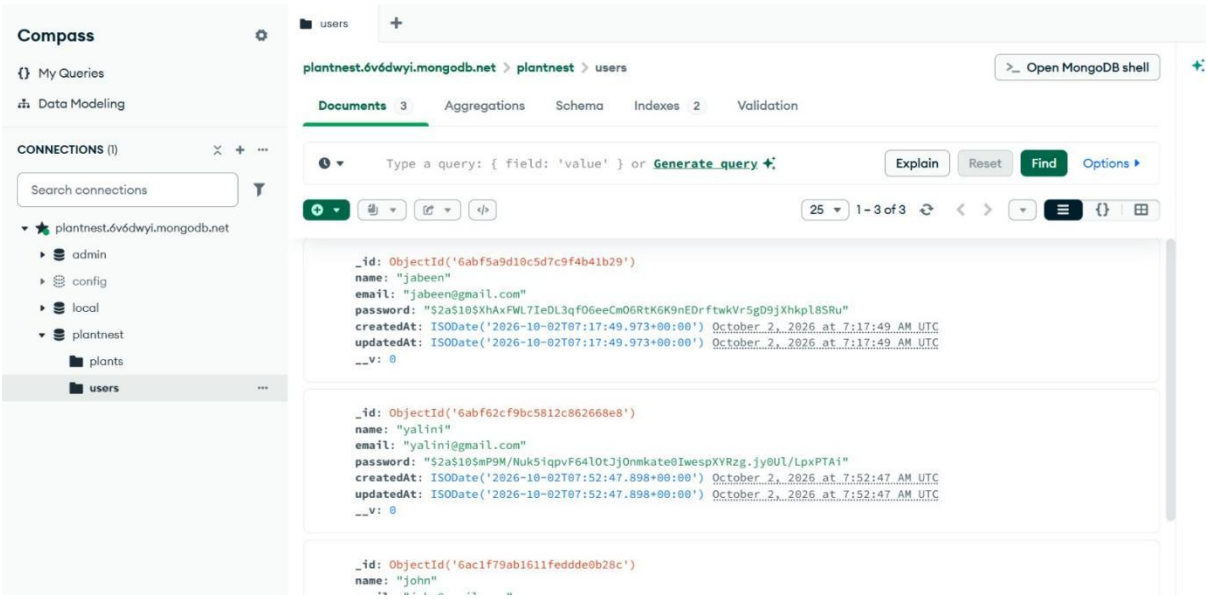


Figure 4.5 – PlantNest MongoDB User Database

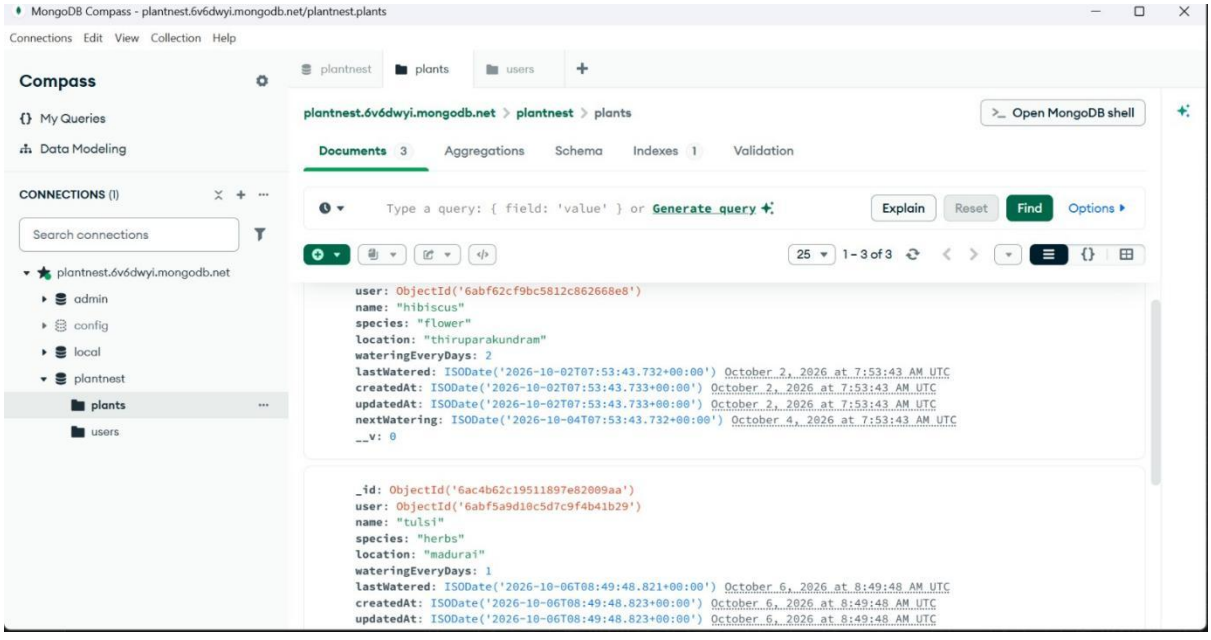


Figure 4.5 – PlantNest MongoDB User Database

Github Link: https://github.com/AJabeen02/plantnest_reminder

CHAPTER 4

CONCLUSION

The **PlantNest – Plant Care and Reminder System** provides a simple and useful web-based solution for managing plants and their watering schedules. The system allows users to register, manage plant records, set watering intervals, and track upcoming watering requirements.

The project uses the MERN Stack, consisting of MongoDB, Express.js, React, and Node.js, to develop the different components of the application. Features such as JWT authentication, plant CRUD operations, automatic next watering date calculation, and daily watering reminders help make plant care more organized.

The system can be further enhanced by adding features such as plant health tracking, weather-based watering suggestions, plant care recommendations, notifications, and plant image identification. Overall, PlantNest provides a convenient digital platform for managing regular plant care and watering activities.

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